

AKSHAR AI

SUMMER INTERNSHIP REPORT

WEEK 4 | 13 JUNE 2026 – 19 JUNE 2026

Project Title: Pneumonia Detection System & Human Behavior Tracking System

DETAILS

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1. Introduction

The integration of artificial intelligence into healthcare and behavioral analysis represents a monumental shift in modern computational biology and human-computer interaction. During the first three weeks of the internship at Akshar AI Ltd, significant progress was made in the conceptualization, preprocessing, and preliminary modeling phases of the Pneumonia Detection System. Utilizing a curated chest X-ray dataset, baseline deep learning models including a Custom Convolutional Neural Network (CNN), MobileNetV2, and ResNet50 were developed, trained, and evaluated. These foundational efforts demonstrated the viability of transfer learning in medical imaging and highlighted critical areas for architectural refinement, particularly the mitigation of overfitting through dataset scaling and robust regularization techniques.

Week 4 marks a pivotal transition in the internship trajectory. It serves as a bridge between the foundational model evaluation of the Pneumonia Detection System and the introduction of an entirely new, highly sophisticated project domain: the Human Behavior Tracking System. This week was strategically divided into two primary operational focuses. The first focus involved a comprehensive review of the deep learning architectures trained in Week 3, followed by meticulous planning for the next phase of training, which involves scaling the dataset incrementally (from 50 samples to 200, 500, 700, and ultimately 1000 images). This iterative scaling approach is designed to empirically analyze the impact of data volume on model generalization, accuracy convergence, and loss minimization.

The second, and arguably more intensive, focus of Week 4 was the initiation of the Hand & Gesture Recognition Module. This module operates as an independent yet integral component of the broader Human Behavior Tracking System. The objective is to engineer a real-time, high-fidelity computer vision pipeline capable of detecting hand presence, localizing 21 distinct 3D landmarks, tracking spatial movements across consecutive video frames, and classifying complex hand gestures with high confidence. This necessitated deep theoretical and practical research into state-of-the-art computer vision frameworks, predominantly MediaPipe and OpenCV, as well as the formulation of standardized JSON output structures required for seamless module integration within the larger tracking ecosystem.

This extensive report chronicles the comprehensive activities, research methodologies, theoretical studies, and technical implementations conducted throughout Week 4. It provides an in-depth analysis of the day-wise progression, the mathematical and computational theories underpinning hand landmark extraction, and the rigorous planning frameworks established for future deep learning experiments. The document is structured to reflect not only the technical milestones achieved but also the profound learning outcomes realized through rigorous research and systematic documentation.

2. Weekly Objectives

To ensure a structured and highly productive week, a series of comprehensive technical and research objectives were defined. These objectives were carefully aligned with the broader goals of both the Pneumonia Detection System and the newly introduced Human Behavior Tracking System. The dual nature of the tasks required a balanced approach to deep learning evaluation and computer vision research.

The primary objectives for Week 4 were formulated as follows:

Comprehensive Project Review: Participate in the weekly review meeting to present, defend, and critically analyze the performance metrics (accuracy, validation loss, convergence speed) of the Custom CNN, MobileNetV2, and ResNet50 models trained during Week 3.

Dataset Scaling Strategy: Formulate a structured, highly systematic experimental design for retraining the aforementioned deep learning models on progressively larger subsets of the Chest X-Ray dataset (200, 500, 700, and 1000 images).

Metric Tracking Framework: Develop a robust, standardized tracking framework (e.g., spreadsheet or database schema) to meticulously record experimental hyperparameters, training durations, accuracy plateaus, and loss trajectories across all future scaling experiments.

Introduction to Human Behavior Tracking: Transition focus toward the Human Behavior Tracking System, specifically taking ownership of the Hand & Gesture Recognition Module. Understand the architectural requirements, integration protocols, and expected JSON output formats.

Computer Vision Research: Conduct extensive literature and technical research on real-time hand detection methodologies. Evaluate the efficacy, latency, and accuracy of frameworks such as Google's MediaPipe, OpenCV, and CVZone for live camera feed processing.

Theoretical Grounding in Landmark Extraction: Study the underlying principles of hand landmark estimation. Understand the topological structure of the 21 3D landmarks (x, y, z coordinates) and how these spatial points translate into recognizable gestures via classification algorithms.

Documentation and Version Control: Consolidate all research notes, experimental plans, architectural diagrams, and weekly reports into professionally formatted documents. Ensure all deliverables are committed and pushed to the centralized GitHub repository with descriptive, granular commit messages.

Day-wise Objective Summary

1	13 Jun 2026	Weekly review of deep learning models & Introduction to Hand Tracking	Actionable feedback on deep learning; requirements for new module
2	15 Jun 2026	Architecture study & JSON standard formulation for Hand Tracking	Understanding of integration protocols and data structures
3	16 Jun 2026	In-depth research on APIs (MediaPipe, OpenCV) for real-time tracking	Technology stack selection for the Gesture Module
4	17 Jun 2026	Planning dataset scaling experiments (200, 500, 700, 1000 images)	Standardized experiment tracking framework

6. Tools and Technologies Used

Deep Learning Framework	TensorFlow / Keras	Referenced for planning the upcoming scaling experiments and model retuning.
Computer Vision Engine	Google MediaPipe	Researched extensively for high-fidelity, real-time hand landmark extraction.
Image Processing	OpenCV	Studied for camera interfacing, image matrix manipulation, and overlay rendering.
High-Level CV Library	CVZone	Evaluated for rapid prototyping of gesture recognition logic.
Data Serialization	JSON	Designed schemas for standardized inter-module communication and data output.
Documentation & Planning	Markdown / Spreadsheets	Used to draft experimental tracking matrices and architectural plans.
Version Control	Git / GitHub	Employed for secure backup, repository management, and version history tracking.
Development Environment	Google Colab / Local IDE	Referenced for the execution environment of future deep learning and CV tasks.

7. Results and Observations

While Week 4 did not produce deployable source code, it generated highly critical architectural and strategic results that dictate the trajectory of the remainder of the internship.

Validation of the MediaPipe Framework: Extensive research confirmed that MediaPipe is the optimal solution for the Hand & Gesture module. Its ability to extract 21 3D landmarks in real-time, even on CPU-constrained environments, addresses the primary latency concerns of the project.

Finalization of the JSON Output Schema: A robust, comprehensive JSON data structure was successfully designed. This schema elegantly captures tracking IDs, handedness probabilities, 3D coordinate arrays, and gesture classifications, completely satisfying the integration requirements of the Human Behavior Tracking System.

Creation of the Experimental Tracking Matrix: A formalized, rigorous framework for logging deep learning experiments was finalized. This matrix ensures that the upcoming dataset scaling experiments (200 to 1000 images) will yield clean, comparable, and statistically valid performance data.

Understanding Kinematic State Estimation: A deep theoretical understanding of how static frame-by-frame landmarks are translated into continuous temporal tracking IDs and semantic gestures was achieved, mitigating the primary algorithmic risk of the new module.

8. Challenges Faced

Context Switching Between Distinct Domains: The most significant challenge of Week 4 was the rapid cognitive context switching required to operate across two highly disparate domains. Shifting from the statistical and probabilistic analysis of CNN performance on static medical imagery (Pneumonia Detection) to the geometric, kinematic, and real-time algorithmic logic of computer vision (Hand Tracking) required intense focus and compartmentalization of concepts.

Designing for Unknown Integrations: Architecting the JSON output schema for the Hand & Gesture module was challenging because the precise specifications of the downstream modules in the Human Behavior Tracking System were not entirely known. Anticipating the data structures that other developers might require (e.g., including both normalized and pixel-space coordinates, providing confidence intervals) necessitated over-engineering the output to ensure maximum compatibility.

Navigating Dense API Documentation: Deeply researching the MediaPipe and OpenCV documentation required navigating complex C++ and Python API references. Understanding the nuances of MediaPipe's two-stage detector and how it handles tracking loss and re-acquisition required careful study of advanced computer vision literature.

9. Learning Outcomes

The research and planning activities of Week 4 yielded profound theoretical and professional learning outcomes:

Advanced Systems Architecture: Developed a strong competency in designing decoupled, modular software components. Learned how to define strict input/output boundaries using JSON serialization to ensure complex systems can be developed in isolation and integrated seamlessly.

Kinematic Computer Vision: Acquired deep theoretical knowledge of state-of-the-art hand tracking methodologies. Understood the transition from simple bounding box object detection to complex, 21-point 3D landmark regression using dual-model pipelines like MediaPipe.

Empirical Experimental Design: Mastered the principles of scientific experimental design within the context of deep learning. Learned how to structure large-scale training runs across varying dataset sizes to isolate variables and definitively quantify model generalization and overfitting.

Algorithmic Translation: Developed the cognitive framework required to translate spatial geometry (Euclidean distances, vector angles between finger joints) into logical heuristic algorithms for gesture classification.

Professional Project Management: Reinforced skills in technical documentation, literature review, and strategic planning, realizing that weeks spent researching and architecting often save months of rewriting code.

10. Conclusion

Week 4 of the Summer Internship was a critical inflection point, serving as a phase of deep research, architectural design, and strategic realignment. Rather than rushing into code implementation, the week was intentionally dedicated to building an unshakable theoretical and planning foundation for both the Pneumonia Detection System and the Human Behavior Tracking System.

For the Pneumonia project, the week culminated in a highly structured, empirical experimental plan. The creation of a rigorous tracking matrix ensures that the upcoming dataset scaling tasks will be executed with scientific precision, ultimately leading to the selection of a highly robust, generalizable medical imaging model.

Simultaneously, the initiation of the Hand & Gesture Recognition Module was met with intense technical scrutiny. By thoroughly evaluating frameworks, selecting MediaPipe, defining the exact JSON integration protocols, and understanding the complex kinematic mathematics of 3D landmark extraction, all structural ambiguity has been eliminated from the project.

The systematic documentation and repository management executed at the week's close have secured all research findings and architectural blueprints. As Week 4 concludes, the transition from theoretical research to practical, aggressive implementation is fully prepared. The groundwork laid this week guarantees that the subsequent development phases will be characterized by high velocity, technical clarity, and seamless systemic integration.